
ELK336-Quantum Computation

HOMEWORK #1

Due 08.03.2024

- The students are obligated to do the homework themselves.
 - The copied homework will not be considered or graded.
 - Homework should be ready before 23.00 on the Ninova in electronic form.
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1- Assume that $\{|a\rangle, |b\rangle, |c\rangle\}$ forms an orthonormal basis in $\mathbb{C}^3$ and

$$|\psi\rangle = \frac{1}{\sqrt{5}}|a\rangle - i\sqrt{\frac{7}{15}}|b\rangle + \frac{1}{\sqrt{3}}|c\rangle \text{ is given.}$$

Calculate the inner product $\langle\psi|\psi\rangle = ?$ Is $|\psi\rangle$ state normalized? **(20 pts)**

2- Given Pauli matrices X, Y, Z ; show that $X^2 = Y^2 = Z^2 = \mathbb{1}$

by using outer product notation. **(15 Pts)**

3- Given $A = i|1\rangle\langle 1| + \frac{\sqrt{3}}{2}|1\rangle\langle 2| + 2|2\rangle\langle 1| - |2\rangle\langle 3|$ with orthonormal basis $\{|1\rangle, |2\rangle, |3\rangle\}$ in $\mathbb{C}^3$,

(a) Find $A^\dagger$ (b) Show if A is unitary or not. **(5pts, 10 pts)**

4- Show that $[A, B]^\dagger = [B^\dagger, A^\dagger]$. **(5pts)**

5- Show that $[\sigma_x \sigma_y, \sigma_z] = 0$. **(15 pts)**

6- Show that two commuting operators A and B share the same eigenstate.

(Hint: Take $A|\psi\rangle = a|\psi\rangle$ and $A|\psi\rangle = |\phi\rangle$) **(15 pts)**

7- Calculate the square root of $A = \begin{bmatrix} 33 & 24 \\ 48 & 57 \end{bmatrix}$ by diagonalization. **(15 pts)**